

Extend uniform subdifferential convergence rates to max-of-smooth weakly convex models

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Website #25 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

Let

$$\phi(x) = \mathbb{E}[f(x, \xi)] + R(x) + \iota_X(x), \quad \phi_S(x) = \frac{1}{m} \sum_{i=1}^m f(x, \xi_i) + R(x) + \iota_X(x),$$

where $\xi_1, \dots, \xi_m$ are i.i.d., R is proper closed convex, $X \subset \mathbb{R}^d$ is closed convex, and ∂ is the Fréchet (regular) subdifferential. The target is a high-probability bound on

$$\sup_{x \in X \cap B(x_0; r)} H(\partial\phi(x), \partial\phi_S(x))$$

for the weakly convex *max-of-smooth* model class, analogous to the convex-composite bound proved in the cited paper.

I prove such an analogue for the stochastic model

$$f(x, \xi) = \max_{1 \leq j \leq N} h_j(x, \xi)$$

with finitely many smooth branches, on an interior ball where the indicator ι_X is inactive, and with a complexity assumption stated as a pseudo-dimension bound for a natural *active-directional-gradient class*. I also give an explicit semialgebraic/polynomial corollary in which that pseudo-dimension is bounded in terms of d and the branch complexity.

Alternative readings considered

- *Max inside the expectation:*

$$f(x, \xi) = \max_{1 \leq j \leq N} h_j(x, \xi), \quad \phi(x) = \mathbb{E}[\max_j h_j(x, \xi)] + R(x) + \iota_X(x).$$

This is the reading I treat; it is the closest stochastic analogue of the paper's setup and is the harder one.

- *Max outside the expectation:*

$$\phi(x) = \max_u \mathbb{E}[h(x, u, \xi)] + R(x) + \iota_X(x).$$

This is easier structurally, but it is not the model I chose.

- The notation $B(x_0; r)$ could mean open or closed ball. For the empirical-process step I work on the closed ball

$$K := \overline{B}(x_0; r),$$

and assume an open neighborhood $U \supset K$ on which all branch regularity holds. If the intended reading was the open ball, my bound applies on every smaller closed ball $\overline{B}(x_0; r') \subset B(x_0; r)$.

Related literature found

- Feng Ruan, *On the Uniform Convergence of Subdifferentials in Stochastic Optimization and Learning*. Classification: **PARTIAL**. This is the paper alluded to in the prompt; it proves a deterministic reduction from selected-subgradient convergence to Hausdorff convergence, and then derives convex-composite rates, but it does not treat max-of-smooth losses. It was useful for orientation only; I do not import its proof. (arxiv.org)
- David Haussler, *Decision Theoretic Generalizations of the PAC Model for Neural Net and Other Learning Applications*. Classification: **TOOL**. I use its high-probability uniform convergence bound for bounded real-valued function classes in terms of empirical L_1 covering numbers. (cis.upenn.edu)
- Akshay Krishnamurthy, Alekh Agarwal, Tzu-Kuo Huang, Hal Daumé III, John Langford, *Active Learning for Cost-Sensitive Classification*. Classification: **TOOL**. I use Lemma 22 there, which states an explicit empirical L_1 covering-number bound in terms of pseudo-dimension (attributed there to Haussler). (jmlr.csail.mit.edu)
- Paul W. Goldberg and Mark R. Jerrum, *Bounding the Vapnik-Chervonenkis Dimension of Concept Classes Parameterized by Real Numbers*. Classification: **TOOL**. I use their theorem to bound the pseudo-dimension of the active-directional-gradient class in the polynomial/semialgebraic branch case. (cs.ox.ac.uk)

Proof sketch

For finite max-of-smooth losses, the sample loss

$$f_\xi(x) := f(x, \xi) = \max_{1 \leq j \leq N} h_j(x, \xi)$$

has a very concrete regular subdifferential at any point x : it is exactly the convex hull of the gradients of the active branches,

$$\partial f_\xi(x) = \text{conv}\{\nabla h_j(x, \xi) : j \in A(x, \xi)\}, \quad A(x, \xi) := \arg \max_j h_j(x, \xi).$$

This part is deterministic and I prove it directly from the definition of the Fréchet subdifferential.

The key move is then *not* to chase arbitrary subgradient selections. Instead, I pass to support functions. For each x and direction $\|v\| \leq 1$, define

$$\psi_{x,v}(\xi) := \max_{j \in A(x, \xi)} \langle \nabla h_j(x, \xi), v \rangle.$$

I prove that for the population risk $g(x) = \mathbb{E}[f(x, \xi)]$ and its empirical version g_S ,

$$H(\partial g(x), \partial g_S(x)) = \sup_{\|v\| \leq 1} \left| \mathbb{E} \psi_{x,v}(\xi) - \frac{1}{m} \sum_{i=1}^m \psi_{x,v}(\xi_i) \right|.$$

So the Hausdorff-distance problem reduces exactly to a scalar empirical-process problem over the class

$$\Psi := \{\psi_{x,v} : x \in K, \|v\| \leq 1\}.$$

Once this reduction is in hand, a standard bounded-class uniform convergence theorem yields a rate in terms of empirical L_1 covering numbers. A pseudo-dimension bound on Ψ then gives the explicit rate

$$O\left(\sqrt{\frac{\text{Pdim}(\Psi) \log m + \log(1/\delta)}{m}}\right).$$

Finally, in the polynomial-branch case, Goldberg–Jerrum gives

$$\text{Pdim}(\Psi) \lesssim d \log(epN),$$

because the subgraph relation $\psi_{x,v}(\xi) \geq t$ is definable by a quantifier-free Boolean formula built from the branch-comparison polynomials and the directional-gradient threshold polynomials.

Main result

Theorem 1 (Uniform Hausdorff bound for finite stochastic max-of-smooth models). *Fix $x_0 \in \mathbb{R}^d$ and $r > 0$, and write $K := \overline{B}(x_0; r)$. Assume there is an open neighborhood $U \supset K$ such that:*

1. $U \subset \text{int } X$;
2. $R \in C^1(U)$;
3. for some finite N ,

$$f(x, \xi) = \max_{1 \leq j \leq N} h_j(x, \xi) \quad (x \in U),$$

where each $(x, \xi) \mapsto h_j(x, \xi)$ and $(x, \xi) \mapsto \nabla_x h_j(x, \xi)$ is jointly measurable;

4. for P -a.e. ξ , each map $x \mapsto h_j(x, \xi)$ is C^1 on U , and

$$\max_{1 \leq j \leq N} \text{Lip}(\nabla_x h_j(\cdot, \xi); U) \leq \beta(\xi)$$

for some measurable $\beta : \Xi \rightarrow [0, \infty)$ with $\mathbb{E}[\beta(\xi)] < \infty$;

5. there is $M < \infty$ such that

$$\max_{1 \leq j \leq N} \sup_{x \in U} \|\nabla_x h_j(x, \xi)\| \leq M \quad \text{for } P\text{-a.e. } \xi;$$

6. if

$$A(x, \xi) := \arg \max_{1 \leq j \leq N} h_j(x, \xi), \quad \psi_{x,v}(\xi) := \max_{j \in A(x, \xi)} \langle \nabla_x h_j(x, \xi), v \rangle \quad (\|v\| \leq 1),$$

and

$$\Psi := \{\psi_{x,v} : x \in K, \|v\| \leq 1\},$$

then $\text{Pdim}(\Psi) \leq V < \infty$.

Define on K

$$\phi(x) = \mathbb{E}[f(x, \xi)] + R(x), \quad \phi_S(x) = \frac{1}{m} \sum_{i=1}^m f(x, \xi_i) + R(x).$$

Then for every $\delta \in (0, 1)$,

$$\Pr \left(\sup_{x \in K} H(\partial \phi(x), \partial \phi_S(x)) \leq 16\sqrt{2} M \sqrt{\frac{V \log(8em) + \log(\frac{4e(V+1)}{\delta})}{m}} \right) \geq 1 - \delta.$$

Corollary 1 (Polynomial-branch corollary). *Under the assumptions of the theorem, assume moreover that $\Xi \subset \mathbb{R}^n$ for some n , and that each branch $h_j(x, \xi)$ is a polynomial in (x, ξ) of degree at most $p \geq 1$. Then there are universal constants $c_0, C > 0$ such that*

$$\text{Pdim}(\Psi) \leq c_0 d \log(epN),$$

and hence

$$\Pr\left(\sup_{x \in K} H(\partial\phi(x), \partial\phi_S(x)) \leq C M \sqrt{\frac{d \log(epN) \log(em) + \log(1/\delta)}{m}}\right) \geq 1 - \delta.$$

Proof

For brevity write

$$g(x) := \mathbb{E}[f(x, \xi)], \quad g_S(x) := \frac{1}{m} \sum_{i=1}^m f(x, \xi_i),$$

so that on $K \subset U \subset \text{int } X$,

$$\phi = g + R, \quad \phi_S = g_S + R,$$

because $\iota_X \equiv 0$ on U .

Lemma 1 (Pointwise max-of-smooth subdifferential formula). *Fix ξ and define*

$$f_\xi(x) := \max_{1 \leq j \leq N} h_j(x, \xi).$$

Then for every $x \in U$,

$$\partial f_\xi(x) = \text{conv}\{\nabla_x h_j(x, \xi) : j \in A(x, \xi)\}.$$

Proof. Fix $x \in U$, abbreviate $A := A(x, \xi)$, and set

$$C := \text{conv}\{\nabla_x h_j(x, \xi) : j \in A\}.$$

We first show $C \subset \partial f_\xi(x)$. Pick any $y \in C$, say

$$y = \sum_{j \in A} \lambda_j \nabla_x h_j(x, \xi), \quad \lambda_j \geq 0, \quad \sum_{j \in A} \lambda_j = 1.$$

For any $h \in \mathbb{R}^d$ with $x + h \in U$, the gradient-Lipschitz assumption gives

$$h_j(x + h, \xi) \geq h_j(x, \xi) + \langle \nabla_x h_j(x, \xi), h \rangle - \frac{\beta(\xi)}{2} \|h\|^2 \quad (j = 1, \dots, N).$$

For $j \in A$, $h_j(x, \xi) = f_\xi(x)$. Hence

$$f_\xi(x + h) \geq \max_{j \in A} \left(f_\xi(x) + \langle \nabla_x h_j(x, \xi), h \rangle - \frac{\beta(\xi)}{2} \|h\|^2 \right).$$

A maximum dominates every convex combination, so

$$f_\xi(x + h) \geq f_\xi(x) + \langle y, h \rangle - \frac{\beta(\xi)}{2} \|h\|^2.$$

Therefore

$$\frac{f_\xi(x+h) - f_\xi(x) - \langle y, h \rangle}{\|h\|} \geq -\frac{\beta(\xi)}{2} \|h\| \rightarrow 0 \quad (h \rightarrow 0),$$

which is exactly $y \in \partial f_\xi(x)$.

We next show the reverse inclusion. Let $y \in \partial f_\xi(x)$. By the definition of the Fréchet subdifferential, for every $v \in \mathbb{R}^d$,

$$\langle y, v \rangle \leq \liminf_{t \downarrow 0} \frac{f_\xi(x+tv) - f_\xi(x)}{t}.$$

So it suffices to compute the directional derivative of f_ξ .

We claim

$$f'_\xi(x; v) = \max_{j \in A} \langle \nabla_x h_j(x, \xi), v \rangle.$$

For the lower bound, fix $j \in A$. Then

$$f_\xi(x+tv) \geq h_j(x+tv, \xi) = h_j(x, \xi) + t \langle \nabla_x h_j(x, \xi), v \rangle + o(t),$$

hence

$$\liminf_{t \downarrow 0} \frac{f_\xi(x+tv) - f_\xi(x)}{t} \geq \langle \nabla_x h_j(x, \xi), v \rangle.$$

Taking the maximum over $j \in A$ gives

$$f'_\xi(x; v) \geq \max_{j \in A} \langle \nabla_x h_j(x, \xi), v \rangle.$$

For the upper bound, take any sequence $t_\nu \downarrow 0$, and for each ν choose $j_\nu \in \{1, \dots, N\}$ such that

$$f_\xi(x+t_\nu v) = h_{j_\nu}(x+t_\nu v, \xi).$$

Since $\{1, \dots, N\}$ is finite, some subsequence has $j_\nu \equiv j_*$. If $j_* \notin A$, then

$$h_{j_*}(x, \xi) < f_\xi(x),$$

so

$$\frac{h_{j_*}(x+t_\nu v, \xi) - f_\xi(x)}{t_\nu} = \frac{h_{j_*}(x+t_\nu v, \xi) - h_{j_*}(x, \xi)}{t_\nu} + \frac{h_{j_*}(x, \xi) - f_\xi(x)}{t_\nu} \rightarrow -\infty,$$

contradicting the fact that the left-hand side equals

$$\frac{f_\xi(x+t_\nu v) - f_\xi(x)}{t_\nu},$$

which is bounded below because f_ξ is M -Lipschitz on U . Hence $j_* \in A$. Along that subsequence,

$$\limsup_{\nu \rightarrow \infty} \frac{f_\xi(x+t_\nu v) - f_\xi(x)}{t_\nu} = \langle \nabla_x h_{j_*}(x, \xi), v \rangle \leq \max_{j \in A} \langle \nabla_x h_j(x, \xi), v \rangle.$$

Since the sequence $t_\nu \downarrow 0$ was arbitrary, the upper bound follows.

Thus, for all v ,

$$\langle y, v \rangle \leq \max_{z \in C} \langle z, v \rangle.$$

If $y \notin C$, the finite-dimensional separation theorem gives some v with

$$\langle y, v \rangle > \sup_{z \in C} \langle z, v \rangle,$$

a contradiction. Therefore $y \in C$. □

Lemma 2 (Population and empirical subdifferential formulas). *For $x \in K$, define*

$$C(x, \xi) := \text{conv}\{\nabla_x h_j(x, \xi) : j \in A(x, \xi)\}.$$

Also define

$$\mathcal{C}(x) := \text{cl}\{\mathbb{E}[\zeta(\xi)] : \zeta \text{ measurable and } \zeta(\xi) \in C(x, \xi) \text{ a.s.}\}.$$

Then for every $x \in K$,

$$\partial g(x) = \mathcal{C}(x), \quad \partial g_S(x) = \frac{1}{m} \sum_{i=1}^m C(x, \xi_i).$$

Moreover, for every $x \in K$ and $v \in \mathbb{R}^d$,

$$g'(x; v) = \mathbb{E}[\psi_{x,v}(\xi)], \quad g'_S(x; v) = \frac{1}{m} \sum_{i=1}^m \psi_{x,v}(\xi_i).$$

Proof. We begin with the population statement.

Step 1: $\mathcal{C}(x) \subset \partial g(x)$. Take any measurable selection ζ with $\zeta(\xi) \in C(x, \xi)$ a.s. By the previous lemma, for each ξ and each h with $x + h \in U$,

$$f(x + h, \xi) \geq f(x, \xi) + \langle \zeta(\xi), h \rangle - \frac{\beta(\xi)}{2} \|h\|^2.$$

Integrating gives

$$g(x + h) \geq g(x) + \langle \mathbb{E}[\zeta(\xi)], h \rangle - \frac{\mathbb{E}[\beta(\xi)]}{2} \|h\|^2.$$

Hence $\mathbb{E}[\zeta(\xi)] \in \partial g(x)$. Since $\partial g(x)$ is closed, $\mathcal{C}(x) \subset \partial g(x)$.

Step 2: directional derivative formula for g . Fix $x \in K$ and $v \in \mathbb{R}^d$. For each ξ , the previous lemma gives

$$f'(x, \xi; v) = \psi_{x,v}(\xi).$$

Also, since each branch gradient is bounded by M , every branch is M -Lipschitz on U , hence so is $f(\cdot, \xi)$. Therefore

$$\left| \frac{f(x + tv, \xi) - f(x, \xi)}{t} \right| \leq M \|v\|$$

for all sufficiently small $t \neq 0$. Dominated convergence yields

$$g'(x; v) = \mathbb{E}[f'(x, \xi; v)] = \mathbb{E}[\psi_{x,v}(\xi)].$$

Step 3: support function of $\mathcal{C}(x)$. We claim

$$\sigma_{\mathcal{C}(x)}(v) = \mathbb{E}[\psi_{x,v}(\xi)].$$

Indeed, if $\zeta(\xi) \in C(x, \xi)$ a.s., then

$$\langle \mathbb{E}[\zeta(\xi)], v \rangle = \mathbb{E}[\langle \zeta(\xi), v \rangle] \leq \mathbb{E}[\psi_{x,v}(\xi)].$$

Taking the supremum over all selections proves

$$\sigma_{\mathcal{C}(x)}(v) \leq \mathbb{E}[\psi_{x,v}(\xi)].$$

For the reverse inequality, choose the lexicographically smallest active directional maximizer:

$$j_*(x, v, \xi) := \min \arg \max_{j \in A(x, \xi)} \langle \nabla_x h_j(x, \xi), v \rangle.$$

Because $N < \infty$ and all relevant maps are measurable, $\xi \mapsto j_*(x, v, \xi)$ is measurable. Set

$$\zeta_*(\xi) := \nabla_x h_{j_*(x, v, \xi)}(x, \xi) \in C(x, \xi).$$

Then

$$\langle \mathbb{E}[\zeta_*(\xi)], v \rangle = \mathbb{E}[\psi_{x, v}(\xi)],$$

so $\sigma_{\mathcal{C}(x)}(v) \geq \mathbb{E}[\psi_{x, v}(\xi)]$. The claim follows.

Step 4: $\partial g(x) \subset \mathcal{C}(x)$. Let $y \in \partial g(x)$. Then for every $v \in \mathbb{R}^d$,

$$\langle y, v \rangle \leq g'(x; v) = \mathbb{E}[\psi_{x, v}(\xi)] = \sigma_{\mathcal{C}(x)}(v).$$

If $y \notin \mathcal{C}(x)$, the separation theorem yields a v with

$$\langle y, v \rangle > \sigma_{\mathcal{C}(x)}(v),$$

contradiction. Hence $y \in \mathcal{C}(x)$, proving $\partial g(x) = \mathcal{C}(x)$.

The empirical statement is identical, with the expectation replaced by the finite average. Concretely,

$$\partial g_S(x) = \frac{1}{m} \sum_{i=1}^m C(x, \xi_i), \quad g'_S(x; v) = \frac{1}{m} \sum_{i=1}^m \psi_{x, v}(\xi_i),$$

because one repeats Steps 1–4 with the probability measure replaced by the empirical measure P_m . $\square$

Lemma 3 (Hausdorff distance equals a uniform empirical process). *For every $x \in K$,*

$$H(\partial \phi(x), \partial \phi_S(x)) = \sup_{\|v\| \leq 1} \left| \mathbb{E}[\psi_{x, v}(\xi)] - \frac{1}{m} \sum_{i=1}^m \psi_{x, v}(\xi_i) \right|.$$

Consequently,

$$\sup_{x \in K} H(\partial \phi(x), \partial \phi_S(x)) = \sup_{\psi \in \Psi} \left| \mathbb{E}[\psi(\xi)] - \frac{1}{m} \sum_{i=1}^m \psi(\xi_i) \right|.$$

Proof. Because $R \in C^1(U)$, we have

$$\partial \phi(x) = \partial g(x) + \nabla R(x), \quad \partial \phi_S(x) = \partial g_S(x) + \nabla R(x).$$

Hence translation invariance of the Hausdorff distance gives

$$H(\partial \phi(x), \partial \phi_S(x)) = H(\partial g(x), \partial g_S(x)).$$

Now let $A, B \subset \mathbb{R}^d$ be nonempty compact convex sets. I claim

$$H(A, B) = \sup_{\|v\| \leq 1} |\sigma_A(v) - \sigma_B(v)|.$$

Indeed, for any $a \in A$,

$$d(a, B) = \sup_{\|v\| \leq 1} (\langle a, v \rangle - \sigma_B(v)).$$

The inequality “ $\leq$ ” holds because $\langle a, v \rangle - \sigma_B(v) \leq \|a - b\|$ for every $b \in B$ and every $\|v\| \leq 1$. For the reverse inequality, choose a nearest point $b_* \in B$ to a ; if $a \notin B$, take $v = (a - b_*)/\|a - b_*\|$, and if $a \in B$, take $v = 0$. This gives equality. Taking the supremum over $a \in A$ yields

$$\sup_{a \in A} d(a, B) = \sup_{\|v\| \leq 1} (\sigma_A(v) - \sigma_B(v)).$$

Exchanging A and B and using the definition of H proves the claim.

Applying this with $A = \partial g(x)$, $B = \partial g_S(x)$, and using the previous lemma,

$$\sigma_{\partial g(x)}(v) = g'(x; v) = \mathbb{E}[\psi_{x,v}(\xi)], \quad \sigma_{\partial g_S(x)}(v) = g'_S(x; v) = \frac{1}{m} \sum_{i=1}^m \psi_{x,v}(\xi_i),$$

we obtain the displayed formula. □

Proof of the theorem. By the previous lemma, it remains to bound

$$\sup_{\psi \in \Psi} \left| \mathbb{E}[\psi(\xi)] - \frac{1}{m} \sum_{i=1}^m \psi(\xi_i) \right|.$$

Define the rescaled class

$$\bar{\Psi} := \left\{ \bar{\psi} := \frac{\psi + M}{2M} : \psi \in \Psi \right\}.$$

Then every $\bar{\psi} \in \bar{\Psi}$ takes values in $[0, 1]$, and

$$\text{Pdim}(\bar{\Psi}) = \text{Pdim}(\Psi) \leq V.$$

We now invoke two external results.

Hypothesis audit for Invoked Result 1 (Haussler/Pollard covering inequality). The parameter set for $\bar{\Psi}$ is

$$T := K \times \{v \in \mathbb{R}^d : \|v\| \leq 1\},$$

which is compact metric because K is compact. The map

$$(\xi, (x, v)) \mapsto \bar{\psi}_{x,v}(\xi)$$

is jointly measurable, since it is built from finitely many jointly measurable branch values and gradients. Therefore $\bar{\Psi}$ is permissible in Haussler’s sense. Also, the sample is i.i.d. and all class members are $[0, 1]$ -valued.

Hypothesis audit for Invoked Result 2 (pseudo-dimension $\Rightarrow$ empirical L_1 cover). We already assumed $\text{Pdim}(\bar{\Psi}) \leq V$.

Applying Invoked Result 1 and then Invoked Result 2 gives, for every $\eta > 0$,

$$\Pr \left(\sup_{\bar{\psi} \in \bar{\Psi}} \left| \frac{1}{m} \sum_{i=1}^m \bar{\psi}(\xi_i) - \mathbb{E}[\bar{\psi}(\xi)] \right| > \eta \right) \leq 4e(V+1) \left(\frac{32e}{\eta} \right)^V e^{-m\eta^2/128}.$$

Choose

$$\eta := \sqrt{\frac{128}{m} \left(V \log(8em) + \log\left(\frac{4e(V+1)}{\delta}\right) \right)}.$$

Since $\log(4e(V+1)/\delta) \geq \log(4e) > 1$, we have $\eta \geq 4/\sqrt{m}$, hence

$$\frac{32e}{\eta} \leq 8e\sqrt{m} \leq 8em \quad (m \geq 1).$$

Therefore

$$4e(V+1) \left(\frac{32e}{\eta} \right)^V e^{-m\eta^2/128} \leq 4e(V+1) e^{V \log(8em)} e^{-V \log(8em)} e^{-\log(4e(V+1)/\delta)} = \delta.$$

So, with probability at least $1 - \delta$,

$$\sup_{\bar{\psi} \in \bar{\Psi}} \left| \frac{1}{m} \sum_{i=1}^m \bar{\psi}(\xi_i) - \mathbb{E}[\bar{\psi}(\xi)] \right| \leq \eta.$$

Undoing the affine rescaling,

$$\sup_{\psi \in \Psi} \left| \frac{1}{m} \sum_{i=1}^m \psi(\xi_i) - \mathbb{E}[\psi(\xi)] \right| \leq 2M\eta = 16\sqrt{2} M \sqrt{\frac{V \log(8em) + \log\left(\frac{4e(V+1)}{\delta}\right)}{m}}.$$

The lemma above identifies the left-hand side with

$$\sup_{x \in K} H(\partial\phi(x), \partial\phi_S(x)),$$

which proves the theorem. $\square$

Proof of the corollary. For pseudo-dimension, it suffices to bound the VC dimension of the subgraph class

$$\mathcal{S} := \{ \{(\xi, t) \in \Xi \times \mathbb{R} : \psi_{x,v}(\xi) \geq t\} : x \in K, \|v\| \leq 1 \}.$$

Fix (x, v) . Then

$$\psi_{x,v}(\xi) \geq t$$

holds iff there exists $j \in \{1, \dots, N\}$ such that

$$h_j(x, \xi) \geq h_k(x, \xi) \quad \forall k = 1, \dots, N, \quad \langle \nabla_x h_j(x, \xi), v \rangle - t \geq 0.$$

Thus membership of (ξ, t) in the subgraph of $\psi_{x,v}$ is expressible by a quantifier-free Boolean formula with:

- concept parameters $(x, v) \in \mathbb{R}^{2d}$;
- instance variables $(\xi, t) \in \mathbb{R}^{n+1}$;
- $O(N^2)$ atomic polynomial inequalities;
- polynomial degree $O(p)$.

Hypothesis audit for Invoked Result 3 (Goldberg–Jerrum). The concept class here is precisely the subgraph class of Ψ ; the number of concept parameters is $2d$; the formula is quantifier-free; the number of atomic predicates is bounded by a constant multiple of N^2 ; and each atomic predicate is polynomial of degree at most p . Restricting v to the unit ball only decreases VC dimension.

Therefore Invoked Result 3 yields

$$\text{Pdim}(\Psi) = \text{VCdim}(\mathcal{S}) \leq c_0 d \log(epN)$$

for some universal constant c_0 . Plugging this into the theorem proves the displayed rate with another universal constant C . $\square$

Gap to the full statement

What remains open in my manuscript relative to the full prompt is precise and limited:

1. I treat the finite-branch stochastic max-of-smooth model

$$f(x, \xi) = \max_{1 \leq j \leq N} h_j(x, \xi),$$

not an arbitrary compact/infinite branch family.

2. I work on an interior ball $K \subset \text{int } X$ and assume $R \in C^1$ on a neighborhood of K . I do *not* prove the extension to nonsmooth convex R or to points where the indicator ι_X contributes nontrivial normal cones.
3. The concentration step is for a bounded active-directional-gradient class Ψ . I do not push the argument to a subexponential-envelope Bernstein bound of the paper’s more refined

$$\max\{\Delta, \Delta^2\}$$

type.

4. My complexity measure is $\text{Pdim}(\Psi)$, with a semialgebraic corollary $\text{Pdim}(\Psi) \lesssim d \log(epN)$. I do not derive a decomposition stated purely in terms of VC dimensions of the *primitive* branch-comparison threshold families.

Self red-team

- **Counterexample attempt 1: ties.** I tested the samplewise formula on

$$f_\xi(x) = \max\{x, -x\} = |x|.$$

At $x = 0$, the active set is $\{1, 2\}$, the active-gradient hull is $[-1, 1]$, and that is exactly the regular subdifferential. So the key lemma survives the worst possible tie.

- **Counterexample attempt 2: stochastic hinge/CDF model.** For

$$f(x, \xi) = \max\{0, x - \xi\},$$

the class Ψ reduces to threshold indicators $\mathbf{1}_{\{\xi \leq x\}}$. Then my reduction gives the empirical-CDF discrepancy, which is the right object. So the theorem specializes to the expected kind of rate.

- **Counterexample attempt 3: boundary of X .** Take $X = [0, \infty)$, $R \equiv 0$, $f \equiv 0$. Then at $x = 0$,

$$\partial \iota_X(0) = N_X(0) = (-\infty, 0],$$

which is unbounded, so the Hausdorff metric in the prompt (defined for nonempty compact sets) is not even available. This confirms that some interiority restriction is genuinely necessary, not cosmetic.

- **Most plausible referee objection 1:** “You used an interchange rule $\partial \mathbb{E}f = \mathbb{E}\partial f$ without justification.” *Response:* I did not use any black-box interchange theorem. I proved both inclusions directly: expectations of measurable active-hull selections are regular subgradients by integrating quadratic lower estimates, and the reverse inclusion comes from the support-function identity for directional derivatives.
- **Most plausible referee objection 2:** “The measurable support-maximizing selection may fail.” *Response:* Finite branching is exactly why it works. I use the lexicographically smallest index in a finite argmax, which is measurable.
- **Most plausible referee objection 3:** “The empirical-process bound is being applied to a nonmeasurable supremum.” *Response:* The class is indexed by the compact metric set $K \times \overline{B}(0; 1)$, and the indexing map is jointly measurable, hence permissible in Haussler’s sense; this is precisely the measurability condition required by the invoked concentration theorem.
- **Boundary cases checked.**
 - $N = 1$: then $\partial f(x, \xi) = \{\nabla h_1(x, \xi)\}$, and the theorem reduces to uniform convergence of gradients.
 - Deterministic case: if $f(\cdot, \xi)$ is a.s. constant in ξ , then $g_S = g$ and the left-hand side is 0.
 - Heavy-tie case: already checked with $|x|$.

Ideas tried

1. **Use Ruan’s deterministic reduction via selected subgradients.** Idea: choose $g(x) \in \partial \phi(x)$, $g_S(x) \in \partial \phi_S(x)$, prove

$$\sup_x \|g(x) - g_S(x)\|$$

is small, then invoke the deterministic reduction. Obstruction: for max-of-smooth losses, any canonical active-index selection is violently unstable near tie manifolds. The toy model $|x| = \max\{x, -x\}$ shows the selected gradient jumps at $x = 0$, so a naive covering/Lipschitz argument collapses. Next step if pursued: use *direction-dependent* selections and then optimize over directions; this is essentially what the successful support-function route does.

2. **Treat $\psi_{x,v}$ as Lipschitz in (x, v) and use a net on parameter space.** Idea: cover $K \times \overline{B}(0; 1)$ by a finite Euclidean net and use Bernstein/union bounds. Obstruction: $(x, v) \mapsto \psi_{x,v}(\xi)$ is generally not Lipschitz in x because the active set can switch discontinuously. Again $|x|$ is the minimal counterexample. Next step if pursued: combine a parameter net with a margin condition ruling out near-ties. I did not assume such a margin because it would be substantially less general than the theorem I obtained.

3. **Work directly with support functions of the subdifferentials.** Idea: identify $\partial f_\xi(x)$ as the active-gradient hull, pass to support functions, and reduce the Hausdorff distance to a scalar empirical process over Ψ . Outcome: this succeeded and produced the theorem.
4. **Extend the result to nonsmooth convex R and boundary points of X .** Idea: combine the stochastic part handled here with a convex sum rule / directional-derivative support formula for $R + \iota_X$. Obstruction: I did not fully audit the needed regular-subdifferential sum rule and compactness of the resulting sets at boundary points, and the prompt’s Hausdorff metric is only defined for compact subdifferentials. Next step if pursued: localize to regions where $\partial R(x) + N_X(x)$ is compact and invoke a fully audited convex-analytic sum rule.

Invoked results

1. **Uniform convergence from empirical L_1 covering numbers.** *Statement.* Let $\mathcal{F}$ be a permissible class of measurable functions $Z \rightarrow [0, M]$, and let $Z_1, \dots, Z_m$ be i.i.d. Then for every $\varepsilon > 0$,

$$\Pr\left(\sup_{f \in \mathcal{F}} \left| \frac{1}{m} \sum_{i=1}^m f(Z_i) - \mathbb{E}[f(Z)] \right| > \varepsilon\right) \leq 4 \mathbb{E}[N_{L_1(P_m)}(\varepsilon/16, \mathcal{F}, \{Z_i\}_{i=1}^m)] e^{-m\varepsilon^2/(128M^2)}.$$

Classification: T00L. Citation: Haussler (1992), Corollary 1, itself credited there to Pollard. (cis.upenn.edu)

Hypothesis-match audit. In the proof I apply this to $\mathcal{F} = \overline{\Psi}$. The class is permissible because it is indexed by the compact metric set $K \times \overline{B}(0; 1)$ and the indexing map is jointly measurable. Every member of $\overline{\Psi}$ is $[0, 1]$ -valued by construction. The sample $\xi_1, \dots, \xi_m$ is i.i.d.

2. **Pseudo-dimension controls empirical L_1 covering numbers.** *Statement.* If a real-valued function class $\mathcal{G}$ satisfies $\text{Pdim}(\mathcal{G}) \leq d$, then for every sample $X = (x_1, \dots, x_n)$ and every $\alpha > 0$,

$$N_{L_1(P_n)}(\alpha, \mathcal{G}, X) \leq e(d+1) \left(\frac{2e}{\alpha}\right)^d.$$

Classification: T00L. Citation: Krishnamurthy–Agarwal–Huang–Daumé III–Langford (2019), Lemma 22, stated there as a Haussler covering-number consequence. (jmlr.csail.mit.edu)

Hypothesis-match audit. I apply this to $\mathcal{G} = \overline{\Psi}$. Affine rescaling does not change pseudo-dimension, so $\text{Pdim}(\overline{\Psi}) = \text{Pdim}(\Psi) \leq V$.

3. **Goldberg–Jerrum VC-dimension bound for semialgebraic concept classes.** *Statement.* Consider a concept class whose concepts are represented by k real parameters and whose membership test is expressible by a quantifier-free Boolean formula with s atomic polynomial equalities/inequalities, each of degree at most d_0 . Then the VC dimension is $O(k \log(sd_0))$; more precisely the paper states a bound of the form $2k \log(8ed_0s)$, up to the author’s logarithm convention. Classification: T00L. Citation: Goldberg–Jerrum (1995), Theorem 2.2. (cs.ox.ac.uk)

Hypothesis-match audit. In the corollary, the concept parameters are $(x, v) \in \mathbb{R}^{2d}$, so $k = 2d$. The instances are $(\xi, t) \in \mathbb{R}^{n+1}$. The subgraph condition $\psi_{x,v}(\xi) \geq t$ is a quantifier-free Boolean formula built from the branch-comparison predicates $h_j - h_k \geq 0$ and the directional-gradient threshold predicates $\langle \nabla_x h_j, v \rangle - t \geq 0$. The number of atomic predicates is $O(N^2)$, and their degree is $O(p)$.

Self-confidence

3 — I checked the deterministic reduction and the concentration step carefully, and I could not break the proof; the only reason I am not at 4 is that the argument is research-level and the support-function/Aumann-selection step is subtle enough that a hidden measurability or closure issue is still conceivable.